

WIRING SYSTEMS (1)

Definition - Wiring system is an assembly of parts used in the formation of one or more circuits

- A wiring system is an assembly of parts used in the formation of one or more electric circuits.

- It consists of a cable, its isolation, its mechanical protection (ie sheathing and bracing) and the various accessories such as joint boxes etc.

Factors affecting the choice of a wiring system

Among the most important points to be considered, ^{like any type of} a wiring system is adopted are:

(a) The type of building

- whether permanent, temporary or an extension to an existing building.

(b) Flexibility

- whether the wiring system will allow future extension, alteration or modification if need be.

(c) Installation conditions

- whether the installation is likely to be subjected to adverse conditions such as mechanical damage, moisture, fumes, weather, abnormal or subnormal temperatures, inflammable (or explosive) dust, gas or vapour.

(e) consider any possibility of flammable (explosive) dust, gas or vapour).

(d) Appearance (ie the nature of the finished job).

- whether the building suits a hidden electrical installation, or allows the appearance of a given installation.

(e) Durability

- whether the installation is to last for the lifetime of the building or not.

NB In normal installation conditions, certain systems

have specific 'life' as shown below:

Wiring system	Approximate years (life) in normal circumstances
(i) PVC Cables on cleats	20 yrs
(ii) Steel-Conduit System with PVC cables (concealed work)	Conduit 35 yrs.
(iii) Steel-Conduit with PVC cables (surface work).	Cables 20yrs, Conduit 30yrs.
(iv) Steel trunking system with copper bars	trunking 35yrs, Cables 20yrs
(v) PVC cables	20 yrs.
(vi) LAs cables	20yrs.
(vii) MICS Cables	25yrs.

(f) Cost.

-to Consider the relative costs of the various alternative methods (systems) of wiring in relation to the estimated life of the installation, and the amount of money available for the electrical installation.

NB! it is important to provide the best job for the smallest amount of money while still making the installation safe to use.

Sometimes Cost is also related to the time of installation, in that the initial Cost might be high but save money in labour costs (eg MICS System versus Steel Conduit system).

(g) Safety.

This is an extremely important factor for two reasons:

- (i) to minimize dangers resulting from the installation

conditions

- (ii) Safety in view of the supply and the earthing arrangements available.

Note There are several wiring systems in the market, each one of them available to perform specific duties in specific circumstances. Some are better than others, but the cheapest is not ~~the~~ always the best, neither the dearest system and vice versa.

SHEATHED WIRING SYSTEMS:

- consists of an insulated cdr provided with a sheath which serves in some degree as a means of mech. damage protection. The insulating materials include:

- impregnated paper,
- rubber
- plastic, and
- mineral insulation.

The sheathing materials include:

- lead,
- tough rubber,
- plastics,
- aluminium &
- textile

The various types of wiring systems are named mainly in respect to the mechanical protection used,

PVC (POLYVINYLCHLORIDE SHEATHED) ^(developed in Germany 1937)

This is an all-insulated system in which the cdrs are insulated with pvc and the sheath is also of an insulating material (PVC).

Although much inferior to rubber, especially in

its elastic properties, it's virtually unaffected by chemical actions, oils, many acids, and alkalis. All in all the insulation resistance of pvc is much lower than that of rubber.

Cable fixings (~~clips~~) are tinned brass buckle clips or lead alloy saddles. The clips are fixed to surfaces with brassed steel pins, and are (clips) available in assorted sizes to accommodate different (various) sizes of cables.

The accessories used with PVC system should as far as possible be insulated (i.e. generally made of bakelite and porcelain material).

Generally the pvc system (cable) is available as:

- Single-core (circular),
- two-core (twin flat), with or without earth, and
- three-core with earth.

Advantages of PVC System

- (a) - cheaper and easier to use as it requires no complicated tools during installation. It's therefore the most commonly used system for domestic wiring. It's also time saving in terms of labour.
- (b) - ~~Since~~ Does not require extensive earthing arrangement (in respect to safety against risk of shock). Since there is negligible metal associated with it.
- (c) Is less affected by oils as compared to rubber.

Disadvantages:

- (a) Cannot withstand severe mechanical damage. This adds to the cost of job, especially at

5

- switch and socket-out drops (to skirting level), which may require extra protection eg Conduit pieces etc.
- (b) Has a low thermal capacity, ie softens at temps above 66°C and tend to crack at very low temperatures. This tend to restrict this system in its general application.

TOUGH RUBBER-SHEATHED (TRS)

- is also an all-insulated system in that the insulation is of vulcanised-rubber and the sheath is tough rubber.

- This cable is available in the following forms:

- Single Core (circular section)

- Flat twin,

- Three core, &

- twin-with-ecc (earth).

- A tough rubber sheathed cable may normally be used without further mechanical protection, although it cannot withstand severe mechanical damage. However it is important to avoid bringing the cable into contact with - gas pipes, water pipes, or other metalwork.

- Regarding applications, the cables may be embedded in plaster, ~~run~~ ^{fixed to} under floors, ~~on the~~ ^{fixed to} surfaces or ~~also~~ drawn in conduits. All connection under floors should be accessible and be made in special boxes of adequate capacity. All ends of cables which do not terminate in boxes must be brought to the surface. Like for the PVC systems, all the accessories used with TRS system should be, as far as possible insulated (generally with bakelite material).

Cable fixing is usually by tinned-brass buckle clips fixed to surfaces with brassed steel pins. Again the clips are available in assorted sizes to accommodate different sizes of cables. ~~For dry sites~~ for damp situations, the clips are of 'Ferry' metal and fixed with brass screws.

Advantages of TRS System:

- (a) Cheaper and easier to use as it requires no complicated tools during installations.
- (b) Does not require extensive earthing arrangement (in respect to safety against risk of shock), there being negligible metal associated with the system.
- (c) Is flexible and impervious to water.

Disadvantages

- (a) TRS is weak to withstand severe mech. damage.
- (b) Is affected by oils.
- (c) Has a limit to the ambient temperature in which it can function efficiently. (i.e. tend to harden in high temperature)

P.C.P (POLYCHLOROPRENE SHEATHED)

Neither PVC nor TRS is quite suitable for farm (agricultural) wiring. For this reason the p.c.p is a plastic material developed to withstand the common farm conditions, such as:

- Steam,
- ammonia fumes,
- lactic acid and milk fats,
- Sulphur fumes.
- direct sunlight, ~~and~~
- heat etc.

The p.c.p cable consists of c.dtrs insulated with vulcanised-rubber and sheathed with p.c.p, which is a tough rubber-like substance.

An alternative to P.C.P is the farm wiring cable "actually known by that name". The farm wiring cable has c.dtrs insulated with butyl-rubber and

7

sheathed with PCP. This gives the cable, very good weather and fire resisting qualities.

Both the PCP and farm wiring cables are installed in the same way as TRS or PVC. Where it is desired to space cables from damp or uneven wall, glazed porcelain cleats (Special cleats) are used.

Accessories such as junction boxes, ceiling rose boxes, lampholder boxes etc if used, are made from phenolic resin compound. The cable entries to boxes are sealed by rubber glands and screw bushers. Where the box is in exposed position it should be ~~field~~ filled with a plastic insulating compound.

Advantages of PCP system:

- Is superior to TRS and PVC, in terms of weather and fire resisting qualities.
- Resistant to oils and petrol.

Disadvantages:

- More expensive than either PVC or TRS.

PILC (PAPER-INSULATED, LEAD-COVERED)

PILC is regarded as a wiring system, as it is used for external underground distribution systems as well as for internal distributions in factories and other premises.

The cable is paper-insulated, lead-sheathed and/or served (compounded jute). The paper is impregnated and must be protected against the ingress of moisture. Such protection is usually afforded by means of sealing boxes.

Compared to VRI Cables of equivalent sizes PILC has greater current carrying capacity. Fixing of PILC cable is by cleats. Further protection against mechanical damage is provided by armouring with helical-wound steel wire or tape.

Advantages of PILC System

- Has greater current carrying capacity than VRI Cables and other similar Cables of equivalent sizes.
- Is flexible, neat and cheap, compared to Conduit systems.

Disadvantages

- Special precautions must be observed in handling and installation. (ie Protection against ingress of moisture and when bending the cable).

Major Precaution to be observed:

- (a) Since paper is hygroscopic, the PILC Cable must be protected against possible ingress of moisture; - usually afforded by means of sealing boxes.
- (b) Since the impregnation comprises, mineral oils and other suitable compounds, it is important to ensure that such impregnation is not dried out when operating in high temperatures (above 75°C).
- (c) A lot of care must be taken when bending the cable to avoid distortion or puncturing (breaking) of the paper.